

Acoustic Obstruction Detection for CSF Shunts

Executive Technology Brief

Domain: Acoustics + Ultrasound + Signal Detection

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Shunt obstruction is detected late (symptoms then imaging)

Mechanism: Catheter-integrated acoustic transducer detects obstruction via pulse-echo impedance contrast

Architecture: Catheter-integrated acoustic transducer that emits ultrasonic pulse and detects reflections from obstruction (debris, tissue, air bubble). Signal classifier identifies obstruction type and severity.

Governing Model: Pulse-echo ultrasound detection. Acoustic impedance contrast at obstruction boundary causes reflection; transducer receives echo; classifier identifies obstruction.

Key Equations:

$Z = \rho * c$ [acoustic impedance of CSF in catheter lumen – P-28 obstruction detection]

$R = (Z_2 - Z_1) / (Z_2 + Z_1)$ [reflection at CSF-obstruction interface in P-28 catheter]

$f * \lambda = c$ [wave equation]

$TOF = 2 * d / c$ [round-trip time of flight]

Attenuation: $A(d) = A_0 * \exp(-\alpha * f * d)$ [frequency-dependent attenuation]

Kill Condition: Tissue obstruction acoustic impedance is too similar to CSF for reliable detection (Z ratio < 1.1)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.